



MECH MAINT- BOILER Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2023-24/LMI/504384

Date:21/09/2023(17:38:10)

Subject:Review of LMI “WORKING IN CONFINED SPACE”

Subject: Review of LMI “WORKING IN CONFINED SPACE”

LMI No: LMI/OGN/OPS/SYST/047

The above approved LMI was prepared in the month of Aug. '2021. The LMI is due for revision in the month of Sept '2023. No changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

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On:21/09/2023(17:38:10) **Ref:**SMTTP/MECH MAINT-/2023-24/LMI/504384

Note No:2

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Note No:3

The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

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Note No:4

The subject LMI has been reviewed and the envisaged changes have been incorporated as per reference document- COS-ISO-00-OGN/OPS/SYST/047 Rev. no-1, Aug'23.

The changes carried out in this revised version are relevant to BMD and TMD areas.

Approval may please be accorded by HOP for this revised LMI 'LMI-OGN/OPS/SYST/047 Revision no-1'.

Submitted for review and for further approval please.

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Note No:5

The LMI has been reviewed after 02 years periodicity. Revision done in clause 4.0 (marked in bold and underlined) in attached doc "NEW LMI".

Submitted for approval please

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Note No:6

Noted

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Note No:10

Please review.

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Note No:11

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Note No:13

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May please be approved.

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Note No:14

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Note No:16

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Note No:17

Please review

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Note No:18

Please review.

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Note No:19

Reviewed. No changes proposed.

Put up for further approval.

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Note No:21

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Note No:23

For responsibility assignment in LMI, it is mentioned as "**Respective Maintenance HOD**" for all clauses. Responsibility may be assigned with clarity.

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Note No:24

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Note No:25

Discussed.

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Note No:26

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Note No:27

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Note No:28

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Note No:29

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Note No:30

Pl review.

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Note No:31

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Note No:32

LMI has been reviewed and no changes proposed.

Put up for further approval.

Submitted by:

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Note No:33

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Note No:34

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Note No:35

May please be approved.

Submitted by:

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Note No:36

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SIMHADRI

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

Local Management Instruction – LMI-OGN/OPS/SYST/047

Revision: 1, Dt: September 2023

LMI for Working in Confined Space
at
SIMHADRI SUPER THERMAL POWER STATION

Approved for Implementation by.....
HEAD OF PROJECT (Simhadri)

Date:

MECHANICAL MAINTENANCE- BOILER

TITLE OF LMI:		Working in Confined Space at Station					
LMI No:		LMI-OGN/OPS/SYST/047					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
<i>01</i>	<i>1</i>	<i>23.09.2023</i>	All changes done as per the COS-ISO-00- OGN/OPS/SYST/047 , Rev. No.: 01,Aug.2023 .	<i>DGM-BMD</i>	<i>ALL HODs</i>	<i>GM(O&M)</i>	<i>HOP</i>

N T P C L I M I T E D

SIMHADRI SUPER THERMAL POWER STATION

LMI APPROVAL DOCUMENT

Title of LMI:	Working in Confined Space at Station	
LMI No.:	LMI-OGN/OPS/SYST/047	
Issue No. : 1	Revision: 1	Date : Sept 2023

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Date of Issue			

Working in Confined Space at Station

INDEX

	Page No
1.0 INTRODUCTION	1
2.0 SUPERSEDED DOCUMENTS	1
3.0 DEFINITION	1
4.0 SCOPE	1
5.0 CONFINED SPACE HAZARDS:	2
6.0 ROLES AND ACCOUNTABILITIES	3
7.0 STEPS OF WORKING	12
8.0 REVIEW:	13
ANNEXURE - I	14
ANNEXURE - II	17
ANNEXURE - III	18

Working in Confined Space at Station

1.0 INTRODUCTION

This LMI provides general guidelines on the safe work practices required for Confined Space and to protect personnel from the hazards of entering & performing jobs in Confined spaces. It contains guidance on safe working and various checks to be carried out before start of work. The LMI has incorporated information on protecting personal safety and health from various hazards that may be encountered during confined space work. The checklist is general in nature and hence some of the points may not apply to a particular work. The result of safe working environment is efficient & quality work.

2.0 SUPERSEDED DOCUMENTS

NIL

3.0 DEFINITION

As per OSHA

"By definition, a confined space:

- *Is large enough for an employee to enter fully and perform assigned work;*
- *Is not designed for continuous occupancy by the employee; and*
- *Has a limited or restricted means of entry or exit.*

These spaces may include underground vaults, tanks, storage bins, pits and diked areas, vessels, silos, and other similar areas.

By definition, a permit-required confined space has one or more of these characteristics:

- *Contains or has the potential to contain a hazardous atmosphere.*
- *Contains a material with the potential to engulf someone who enters the space.*
- *Has an internal configuration that might cause an entrant to be trapped or asphyxiated by inwardly converging walls or by a floor that slopes downward and tapers to a smaller cross section; and/or*
- *Contains any other recognized serious safety or health hazards"*

4.0 SCOPE

Confined spaces may include but are not limited to:

- Storage tanks, process vessels, **boilers components**, pressure vessels and silos

- Culverts, Tunnels, Open-topped spaces such as pits or degreasers
- Pipes, sewers, shafts, ducts, and similar structures.

Some places may not be confined space at the start of work but may become confined spaces when work is carried out, or during their construction, fabrication, or subsequent modification.

Based on above, a list of Confined spaces has been prepared and an identification number has been assigned as per Annexure III. This list at Annexure III is not exhaustive and respective HODs can further identify the confined spaces in the plant/ project areas and assign identification number in continuation to it. **Station BMD shall be custodian of LMI for this OGN. Station BMD shall be responsible for confined spaces of Boiler & Aux. only.**

5.0 CONFINED SPACE HAZARDS:

There are various types of hazards that can exist in confined spaces. These hazards can be present in non-confined spaces also, but characteristics of a confined space usually aggravate or intensify many of these hazards. Some of the hazards frequently encountered in the confined spaces are:

5.1 Atmospheric Hazards:

- a. Suffocation (or asphyxiation) hazards — due to oxygen deficiency.
- b. Fire/Explosion hazards — due to presence of flammable gases and vapors, coal dust etc.
- c. Poisoning — due to presence of toxic gases, vapors, or fumes
- d. Oxygen Enrich atmosphere
- e. Hypoxic air
- f. Heat stress- hot and humid

5.2 Non – Atmospheric Hazards: These includes hazards other than atmospheric hazards.

- a. Poor Lighting
- b. Engulfment – Stepping on loosely packed particles, collapsing loosely packed Particles, Inadvertent opening of feed lines to confined space.
- c. Entrapment – Tapering or inwardly sloping and smooth walls
- d. Ergonomic Hazards – Limited Space
- e. Falling from height
- f. Falling Object.

- g. Claustrophobia- Fear from darkness and other problems associated with being in dark, cramped & isolated places.
- h. Disorientation.
- i. Mechanical hazard such as moving or rotating parts.
- j. Electrical Hazards such as improper electrical wiring, wet spaces, poor housekeeping of electrical cables, no provision of grounding etc.
- k. Loss of communication
- l. High Intensity Noise
- m. Drowning – Inadvertent opening of liquid supply line etc.

6.0 Roles & Accountabilities

During entry to confined space for carrying out works, a safe system in the form of written document must be developed and implemented considering the following:

- a. Respective area HODs shall identify and prepare a list of confined spaces in each area including their Hazards and prepare its Standard Maintenance Procedure, Emergency Rescue Plan, Rescue Team etc. Further, a cross functional team in the station shall list out the confined spaces and hazards involved and type of gases to be tested before issuing the permit, as not all confined spaces such as water /steam stored tanks require complete list of gas testing. For some systems such as LDO/HFO, chemical tanks etc., other gases may also be tested and individual check list for different confined space to be made as per Annexure -1. Use of chemicals in confined space - Respective maintenance shall ensure proper PPE are used while using chemicals inside confined space. Separate JSA to be made for same. MSDS of the chemical shall also be available. No flammable chemicals shall be used in confined space.

Responsibility- Respective Maintenance HOD/ CFT (consisting of EIC, Chemistry/Safety)

- b. Identify Entrant (employee/agency's manpower)/ supervisor/ attendant work responsibilities.

Responsibility- Respective Maintenance HOD

- c. Identify and evaluate permit space hazards.

Responsibility- Respective Maintenance HOD/ Head-Operation

-
- d. Establish and implement the means, procedures, and practices to eliminate or **control hazards** necessary for safe permit space entry operations.
- i. Specifying acceptable entry conditions.
 - ii. Isolating the confined space.
 - iii. Providing barriers.
 - iv. Verifying acceptable entry conditions; and
 - v. Purging/ making inert/ flushing or ventilating the permit space.
 - vi. Removal of solid wastes, coal dust, contaminants etc.

Responsibility- Respective Maintenance HOD/ Head-Operation

- e. Arrangements should be made for proper (natural/forced) ventilation inside space with fresh air, free from fumes and contaminants.

Responsibility- Respective Maintenance HOD

- f. Test atmospheric conditions in the permit space before entry operations and monitor the space during work at predefined interval. Initially chemistry department shall ensure the atmospheric hazards are within limit by measuring oxygen, combustible gases or vapors, and toxic gases etc. However, during work, measurement of oxygen level, temperature and other parameters in confined space will be performed at predefined interval by Operation/ Maintenance Department. The limits of gases shall be kept as follows:
- I. O₂: 19.5 % - 23 %
 - II. Combustible gases: Below 10% LEL (lower explosive limit).
 - III. Toxic gas: below permissible exposure limit (PEL)/ threshold limit value (TLV)
Further LEL value and PEL / TLV value of some gases are given in Annexure II.
 - IV. Before entry in confined space, Competent Person (CP) & CFT (cross functional safety team) members will check the space themselves and ensure that space is safe for entry.

Responsibility- Respective Maintenance HOD/ HOD-Opn/HOD-Chemistry/CFST

-
- g. There should be arrangement for adequate illumination inside the confined space. All portable illumination system shall be up to 24 Volt only. (Ref 36(a) of Factories Act 1948).

Responsibility- Respective Maintenance HOD/HOD-EMD

- h. Personal protective equipment in space to be identified. PPE includes Helmet/ Shoe/ Safety Belt/ Suitable Dust Mask/ Breathing Apparatus/ Hand Gloves/ Safety Goggles/ Earmuff or Plugs/ Protective Clothing/ Face Shield/ Torch/ Personnel gas detector etc.

Responsibility- Respective Maintenance HOD/HOD-Safety

- i. Ensure that at least one attendant is stationed outside the permit space for the duration of entry operations.

Responsibility- Respective Maintenance HOD

- j. Implement necessary measures to prevent unauthorized entry such as barricading openings, keeping entry register at entry point for recording movements in confined space, Noticeable cautions sign "Notice - Confined Space Entry Permit Required for Entry." shall be displayed.

Responsibility- Respective Maintenance HOD

- k. Proper means of communication should be identified between Entrant, Supervisor, Attendant, Rescue team member during work.

Responsibility- Respective Maintenance HOD

- l. Proper Coordination of entry operations when employees of more than one employer are working in the permit space.

Responsibility- Respective Maintenance HOD

- m. The following list of information should be displayed outside at confined space entry point:
 - I. Hazards in space,
 - II. List of all people working in the space along with their trade/ work responsibility, Supervisor, Attendant, Rescue team members along with their Contact numbers,
 - III. Contact numbers of other important location such as fire services, control room, First Aid, Ambulance etc.
 - IV. Details of Gas, Chemicals & Equipment being used in space,
 - V. PPE to be worn for entry into space,
 - VI. Readings of all measurements taken in Space and measurements taken during work

Responsibility- Respective Maintenance HOD

- n. Debriefing, housekeeping of confined space before releasing confined space for placing it in service.

Responsibility- Respective Maintenance HOD

- o. Implement appropriate procedures prepared for summoning rescue and emergency services and preventing unauthorized personnel from attempting rescue.

Responsibility- Respective Maintenance HOD

- p. Implement the procedures that any attendant who is required to monitor multiple spaces will follow during an emergency in one or more of those spaces.

Responsibility- Respective Maintenance HOD

6.1 Detection of Hazardous conditions: -

- a. If hazardous conditions are detected during entry, employees must immediately leave the space
- b. Evaluation must be done for the space to determine the cause of the hazardous atmosphere and modify the program as necessary.
- c. When entry to permit spaces is prohibited, take effective measures to prevent unauthorized entry.

Responsibility- Respective Maintenance HOD

6.2 Informing employees and contract workers:

It must be informed to any contractor employees who are hired to enter confined spaces about:

- a. The confined spaces and its entry requirements
- b. Any identified hazards
- c. The previous experience with the space, such as knowledge of hazardous conditions
- d. Precautions or procedures to be followed when in or near confined spaces.

Above information may be in form of Job Safety Analysis (JSA), Hazard Identification and Risk Assessment (HIRA), Toolbox Talk (TBT) etc. When employees of more than one employer are conducting entry operations, the affected employers must coordinate entry operations to ensure that affected employees are appropriately protected from permit space hazards. The employer also must give contractors any other pertinent information regarding hazards and operations in permit spaces and be debriefed at the conclusion of entry operations.

Responsibility- Respective Maintenance HOD

6.3 Training to employees & workers:

Before the initial work assignment begins, proper training for all workers who are required to work in confined spaces. After the training, it must be ensured that the employees have acquired the understanding, knowledge, and skills necessary to safely perform their duties. Additional training is required when:

- a. The job duties change
- b. A change occurs in the permit space program, or the permit space operation presents any new hazard
- c. An employee's job performance shows deficiencies.

In addition to this training, rescue team members also require training in cardiopulmonary resuscitation (CPR) and first aid. It must be certified that this training has been provided. After completion of training, record of employee training must be kept and make it available for inspection by employees and their authorized representatives. The record must include the employee's name, the trainer's signature or initials and dates of the training.

Responsibility- Respective Maintenance HOD/HOD-Safety/CMO

6.4 Authorized entrants:

Authorized entrants are required to:

- a. Know space hazards, including information on the means of exposure such as inhalation or dermal absorption, signs of symptoms and consequences of the exposure
- b. Use appropriate personal protective equipment properly
- c. Maintain communication with attendants as necessary to enable them to monitor the entrant's status and alert the entrant to evacuate when necessary
- d. Exit from the permit space as soon as possible when:
 - i. Ordered by the authorized person
 - ii. He or she recognizes the warning signs or symptoms of exposure
 - iii. A prohibited condition exists
 - iv. An automatic alarm is activated.
- e. Alert the attendant when a prohibited condition exists or when warning signs or symptoms of exposure exist.

Responsibility- Respective Maintenance HOD

6.4.1 Attendant:

The attendant is required to:

- a. Remain outside the permit space during entry operations unless relieved by another authorized attendant
- b. Perform non-entry rescues when specified by the rescue procedure
- c. Know existing and potential hazards, including information on the mode of exposure, signs or symptoms, consequences, and physiological effects.
- d. Maintain communication with and keep an accurate account of those workers entering the permit space.
- e. Order evacuation of the permit space when:
 - i. A prohibited condition exists.
 - ii. A worker shows signs of physiological effects of hazard exposure.
 - iii. An emergency outside the confined space exists; and
 - iv. The attendant cannot effectively and safely perform required duties.
 - v. Summon rescue and other services during an emergency.

- vi Ensure that unauthorized people stay away from permit spaces or exit immediately if they have entered the permit space.
- vii. Inform authorized entrants and the entry supervisor if any unauthorized person enters the permit space; and
- viii Perform no other duties that interfere with the attendant's primary duties.

Responsibility- Respective Maintenance HOD

6.4.2 Entry Supervisor:

Entry supervisors are required to:

1. Know space hazards including information on the mode of exposure, signs or symptoms and consequences.
2. Verify emergency plans and specified entry conditions such as permits, tests, procedures, and equipment before allowing entry.
3. Terminate entry and cancel permits when entry operations are completed or if a new condition exists.
4. Verify that rescue services are available and that the means for summoning them are operable.
5. Take appropriate measures to remove unauthorized entrants; and
6. Ensure that entry operations remain consistent with the entry permit and that acceptable entry conditions are maintained.

II. Responsibility- Respective Maintenance HOD

6.5 Emergencies:

6.5.1 Rescue Personnel

Ensure that responders can respond to an emergency in a timely manner. Rescue service personnel with personal protective and rescue equipment must be ensured, including respirators, and training in how to use it. Rescue service personnel also must receive the authorized entrants training and be trained to perform assigned rescue duties. The standard also requires that all rescuers be trained in first aid and CPR. At a minimum, one rescue team member must be currently certified in first aid and CPR. It must be ensured that practice rescue exercises are performed yearly and that rescue services are provided access to permit spaces so they can practice rescue operations. Rescuers also must be informed of the hazards of the permit space.

Responsibility- Respective Maintenance HOD/HOD-Safety/CMO

6.5.2 Harness and retrieval lines:

Authorized entrants who enter a confined space must wear a full body harness with a retrieval line attached to the center of their backs near shoulder level or above their heads. Also, the employer must ensure that the other end of the retrieval line is attached to a mechanical device or a fixed point outside the permit space. A mechanical device must be available to retrieve someone from vertical type permit spaces more than five feet (1.524 meters) deep.

Responsibility- Respective Maintenance HOD

6.5.3 MSDS:

If an injured entrant is exposed to a substance for which a Material Safety Data Sheet (MSDS) or other similar written information is required to be kept at the worksite, that MSDS or other written information must be made available to the medical facility personnel treating the exposed entrant.

Responsibility- Respective Maintenance HOD

6.6 Additional Precautions for working in confined space:

- 6.6.1 Use multi-Gas Meter and Measure every 4 feet in direction of depth. Monitor oxygen level before entry and during the work.
- 6.6.2 Ensure that the confined space is thoroughly free of all hazardous or corrosive substance such as gases, all kinds of fumes, chemicals, muck and solid wastes, coal dust etc. Also, it must be free from decaying vegetation and animal waste that may release Methane gas.
- 6.6.3 The Coal Dust in confined space shall not be cleaned by broom, compressed air, high pressure water jets as it will result in coal dust becoming air borne and forming explosive mixture (Minimum Explosible Concentration in air: 20 to 100gm/m³). Coal dust in confined space should be removed by suitable vacuum cleaner for the

intended work. In case, coal dust is cleaned by water, then area to be flooded with water to avoid it becoming air borne.

- 6.6.4 It shall be ensured that the confined space is completely isolated from the connected system with blank plates having correct size, material of construction and appropriate pressure rating wherever necessary and separated from all the utility systems such as inert gases, fuel gases, steam, water, and other fluid connections.
- 6.6.5 Ensure that all the moving parts of equipment inside the confined space are adequately guarded and the power supply disconnected with LOTO.
- 6.6.6 Ensure that all the portable electrical equipment used in a confined space are either grounded or double insulated or equipped with earth leakage cut out safety system. In case there is a likelihood of presence of explosive fumes or gases, all the electrical devices used must be of a type approved for such application.
- 6.6.7 Continuous monitoring shall be used when the activity inside the confined space can alter atmospheric conditions or there is a known activity taking place outside the space during the entry that has direct potential to alter the atmospheric conditions inside the space. Periodic monitoring (2 hours interval or as case may be) shall be done in the cases when the condition inside the confined space is not likely to be altered because of the work inside the confined space and/or activities being carried out in the surrounding space
- 6.6.8 According to the nature of work and the risk involved due to presence of toxic gases and fumes, the working spell shall be interrupted by rest periods during which the person shall come out and breathe in open air or take breathing apparatus with them. The maximum allowable spell of work shall be recorded clearly in the written procedure. Maximum working hours of a person inside confined space should be mentioned in written procedure
- 6.6.9 In case of gas cutting and welding in a confined space, ensure that all the houses are free from leaks. Compressed gas bottles/cylinders are forbidden inside a confined space. The gas torches should be lighted outside the confined space and the space must be tested for explosive atmosphere each time before a lighted torch is taken inside. [Ref Section 37.4 of Factories Act 1948]
- 6.6.10 As far as practicable all confined space job area to be planned during day light hour.
- 6.6.11 Whenever combustion type equipment is used in a confined space, ensure that the exhaust gases are vented outside.
- 6.6.12 Size of the entrance/ exit shall be big enough to allow workers wearing all the necessary equipment to climb in and out easily and provide ready access and exit in an emergency.

- 6.6.13 Fume Extractor may be required to extract fumes generated in the space which may not be removed by ventilation.
- 6.6.14 All the openings inside confined space may be covered/ barricaded to prevent falling of men & material during work.

Responsibility- Respective Maintenance HOD/HOD-Operation

7.0 STEPS OF WORKING:

- a. Carry out Initial assessment by considering Physical/ other hazards and atmospheric condition.
- b. Follow the written down Work procedure after detailed Hazard Identification & Risk Assessment or Job Safety Analysis (JSA) in conjunction with Safety department. It must include:
 - I. Isolations
 - II. Hazard identification and means of controlling it.
 - III. Measuring gases and oxygen level
 - IV. Ventilation (natural/ forced)/ Purging.
 - V. Illumination
 - VI. PPE
 - VII. Communication
 - VIII. Earthing of all electrical appliances before its use.
 - IX. Emergency Rescue Plan
- c. Forming team for work & ensure that medical examination of all workers are completed.
- d. Make arrangement for Safe working in accordance with above Safe work procedure.
- e. Fill the check List as per Annexure I and take confined space permit.
- f. Display all information at entry point.
- g. Tool box talk and Briefing to all workers
- h. Ensuring all emergency rescue plan in place before entry into confined space.
- i. Start of Work
- j. Permit shall be cancelled immediately if old hazard reappear/ new hazard appears.
- k. Continuous Gas monitoring such as oxygen, flammable gas/ toxic gas(if any) while working

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- l. Housekeeping and Debriefing after work completion

Responsibility- Respective Maintenance HOD

- m. Closing of work permit

8.0 REVIEW:

The Head of Corporate Operation Services will be responsible for reviewing this document on a 2-yearly basis or as necessary.

ANNEXURE - I
CONFINED SPACE CHECKLIST

Station : Unit No.: Date of Issue :
CSP No.: Valid From : Valid To :

Functional Location equipment:

Maint. Order No.:

Hazards Identified: Atmospheric*(CO/CO₂/H₂S/SO_x/NH₃/ Cl₂/ Coal Dust/ Oxygen Deficient /Steam Vapor/ Others_____

Non-Atmospheric (Entrapment /Engulfment/Ergonomic/ Falling Objects/ Fall from Height/ Electrical/ Others

Entry Required For: Hot Work/ Repair/ Inspection/ Electrical Work/ Cleaning/ Painting/ Others

Special Precautions to be taken for work (If any): _____

The following checklist from S.No. 1 to 20 (ensured by SAP & CP) should be filled before issue of permit and remaining S.No. 21 to 24(ensured by CP) should be filled before entry of personnel inside confined Space. However, Compliance of all points in the checklist shall be ensured by SAP, CP/ Permit Receiver & CFT members.

(Entry shall not be allowed/ Ongoing Work to be stopped if checklist is not filled and signed by all concerned):

S.No.	DESCRIPTION	Yes/ No
1	All Available openings in Space is opened for Ventilation	Yes/ No
2	Entry Point, Exit Point, Rescue Point is tagged and cordoned off for Authorized entry only	Yes/ No
4	Entry Register is available with Attendant for recording movement in Space.	Yes/ No
5	Atmospheric Hazards in Space is within limit (Refer Note 3 & Annexure – I)	Yes/ No
6	Non-Atmospheric Hazards in Space is taken care off	Yes/ No
7	Purging is completed (if applicable)	Yes/ No
8	Oxygen Content in Space (Min. 3 Locations) after purging (if required) and before entry is within limit (>19.5% and < 23%)	Yes/ No
9	Temperature (°C) & Humidity in Space (%) before entry is safe and comfortable	Yes/ No
10	All isolations are rechecked personally by Permit receiver before receiving permit	Yes/ No
11	Illumination adequate and only 24 V power supply is used in Space	Yes/ No
12	Information is displayed outside at entry points containing all detail as per Note 9	Yes/ No

S.No.	DESCRIPTION	Yes/ No
13	Forced ventilation of space with fresh air (free from fumes and contaminant) is maintained before entry and during work	Yes/ No
14	Cross permits issued for works in space	Yes/ No
16	Chemicals to be used inside space	Yes/ No
17	Flammable Liquid or Gas to be used inside space	Yes/ No
18	Equipment (Electrical, Mechanical, Others) to be used inside space	Yes/ No
19	Monitoring Frequency of atmosphere of Space (O ₂ , Temperature, Humidity, Gases) within specified interval.	Yes/ No
20	Emergency Response is in place before entry.	Yes/ No
21	Electrical Equipment inside space is properly earthed before use	Yes/ No
22	Required PPE are worn by all person before entry in space	Yes/ No
23	All person working in Space are trained for working in Confined Space.	Yes/ No
24	Toolbox Talk is given on confined space before entry.	Yes/ No
25	CP/ Permit Receiver & CFT member have checked the space himself and found it safe before giving permission for entry into space.	Yes/ No

Any other details:

Permit Issued To By (CP)	Permit Issued By (SAP)	Permit Verified (CFST Members)
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SAP: Senior Authorised Person,

CP: Competent Person

PI refer notes

Notes:

- 1) Stations shall identify and prepare a list of confined spaces in each area including their Hazards, also prepare its Standard Maintenance Procedure, Emergency Rescue Plan, Rescue Team etc. and mock drill may be carried out at regular intervals.
- 2) *Measuring instrument for combustible gases or vapors or mist/ toxic gases/ explosive dusts to be in place.
- 3) Atmospheric Hazards: Combustible gases or vapors or mist (Safe Limit <10% LEL) / Toxic Gases (Safe Limit is below PEL/ TLV)/ Dust/ Oxygen Deficient (If O₂ <19.5%)/ Oxygen Enrich (If O₂ >23%)/ Others. (CO/CO₂/H₂S/SO₂/Ammonia/ Chlorine/ NO_x/Coal Dust/ Suspended Fly Ash/ Steam Vapors/ Others)

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- 4) The Coal Dust shall not be cleaned by broom, compressed air, high pressure water jets as it will result in coal dust becoming air borne and forming explosive mixture (Minimum Explosible Concentration in air: 20 to 100gm/m³). Coal dust in closed space should be removed by suitable vacuum cleaner for the intended work. In case, coal dust is cleaned by water, then area to be flooded with water to avoid it becoming air borne.
 - 5) Non-Atmospheric Hazards: Entrapment / Engulfment/ Rotating parts in space/ Electrical Hazards in space / Fall from height/Falling objects/ Ergonomic hazard/ Noise/ High temperature/ High humidity etc.
 - 6) If chemicals/ flammable liquid or gas are used in Space, then Material Safety Data Sheet to be displayed at site and adequate precautions to be taken.
 - 7) PPE: Helmet/ Shoe/ Safety Belt/ Suitable Dust Mask/ Breathing Apparatus/ Hand Gloves/ Safety Goggles/ Earmuff or Plugs/ Protective Clothing/ Face Shield/ Torch/ Personnel gas detector/ Others.
 - 8) If any inflammable gas, fume, or dust is likely to be present in such chamber, tank, vat, pit, pipe, flue, or other confined space no lamp or light other than that of flame-proof construction shall be permitted to be used therein.
 - 9) The list of information to be displayed outside at entry point shall include list of all person working in the Space along with their trade/ responsibility, Supervisor, Attendant, Rescue team members along with their Contact numbers, contact numbers of other important location such as fire services, control room etc., Hazards in space, Details of Gas, chemicals & equipments being used in space, PPE to be worn, all measurements taken in Space and as per S.No.: 19.
 - 10) Fume Extractor may be required to extract fumes generated in the space which may not be removed by ventilation.
 - 11) Emergency response includes Emergency rescue plan in place, applicable retrieval devices near entry point, Ambulance nearby Space, First Aid Facility in the Station etc.
 - 12) Proper Means of communication should be established before entry and during work in confined space.
 - 13) The permit to be cancelled immediately, if old hazard reappears or new hazard appears.
 - 14) Spot checking of permit may be done during work.
 - 15) Non-Sparking tools to be used in enclosed space having flammable gases.
 - 16) All the openings inside confined space may be covered/ barricaded to prevent falling of men & material during work.
 - 17) The above checklist is not intended to be exhaustive but includes many of the essential elements to help prepare a safe system of work. Station may further add check points as per specific work.
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ANNEXURE - II

Permissible Exposure Limits

Substance	Permissible Limits of Exposure			
	Time - Weighted Average Concentration (TWA) (8 Hrs.)		Short - Term Exposure Limit (STEL) (15min)*	
	PPM	Mg./M ³	PPM	Mg./M ³
Ammonia	25	18	35	25
Benzene (HC)**	0.5	1.5	25	7.5
Carbon Monoxide	20	23	100	117
Chlorine	-	-	0.5	1.5
Hydrogen Sulphide	5	7	10	14
Nitric Oxide	25	30	-	-
Nitrogen Dioxide	3	6	5	10
Sulphur Dioxide	2	5	5	10
Welding Fumes	-	5	-	-

Source: Standard For Working In Confined Spaces (CC-SAFETY-STD/COMMON/CS/R0 dtd Sep. 2018)

Lower Explosion Limit (LEL) and Upper Explosion Limit (UEL) of some flammable gases (At atmospheric temperature and pressure)

Substance	LEL in % by volume of air	10% of LEL in % by volume of air	UEL in % by volume of air	Autoignition Temperature	Remarks
Acetylene	2.5	0.25	100	305 °C	Use of D.A.
Carbon monoxide	12.5	1.25	74	609 °C	Mills, Ducts
Hydrogen	4	0.4	75	500–571 °C	
Hydrogen Sulfide	4.3	0.43	46		Likely presence in Low Temp. region of Boiler
LPG (Liquified Petroleum Gas)	1.9	0.19	9.5	466.1	MSDS of LPG of HPCL

Source: From Web

ANNEXURE - III

List of Confined Spaces with Identification Number

Identification Number	Location	Equipment / Area
A1	Ash Handling	Bottom Ash Hopper
A2	Ash Handling	Ash Silo
A3	Ash Handling	Buffer Hopper / Air Lock Tank
B1	Boiler	<u>Boiler wind box</u> (As per description of SAP)
B2	Boiler	Boiler Drum
B3	Boiler	Bottom Ring Header
B4	Boiler	Coal Bunker (As per description of SAP)
B5	Boiler	Coal Mill (As per description of SAP)
B6	Boiler	Air & Flue Gas Ducts (As per description of SAP)
B7	Boiler	ESP (As per description of SAP)
B8	Boiler	Auxiliary boiler combustion chamber
B9	Boiler	Auxiliary boiler Drum
B10	Boiler	Auxiliary boiler Mud Drum
B11	Boiler	Auxiliary boiler Duct
B12	Boiler	ECW Storage Tank
B13	FO system	HFO tank
B14	FO system	LDO tank
T1	Turbine	Deaerator
T2	Turbine	Condenser
T3	Turbine	Pits such as CEP Pit
T4	Turbine	Flash Tank
T5	Turbine	HP & LP Heater
T6	CompressorHouse	Air Receiver Tanks
<u>T7</u>	<u>Turbine</u>	<u>Main oil tank</u>
<u>T8</u>	<u>Turbine</u>	<u>Control fluid tank</u>
<u>T9</u>	<u>Turbine</u>	<u>Waste oil tank</u>
<u>T10</u>	<u>Turbine</u>	<u>Clean oil tank</u>
<u>T11</u>	<u>Turbine</u>	<u>Dirty oil tank</u>
OF1	OFS	CW Ducts

D1	DM Plant	Ion Exchange Vessel
D2	DM Plant	Water Storage Tanks
D3	DM Plant	Chemical Storage Tanks
D4	DM Plant	Degasser Storage Tank
C1	CHP	Coal Transfer chutes/ Hoppers/ Silos
C2	CHP	Track Hoppers

[illegible]